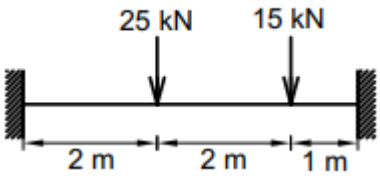
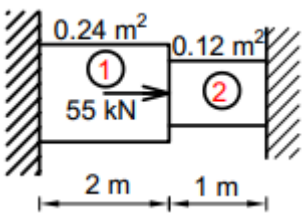
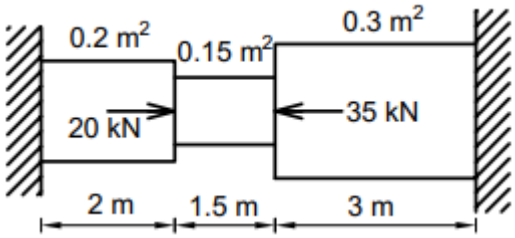
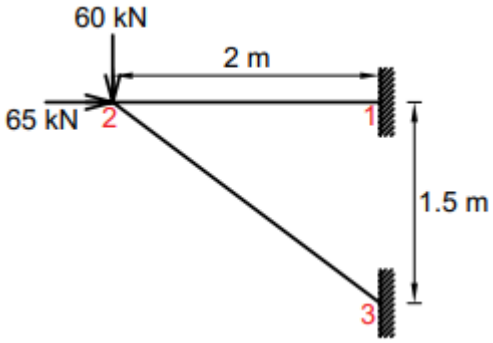
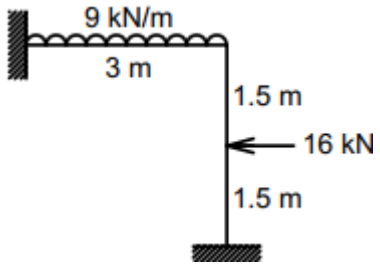
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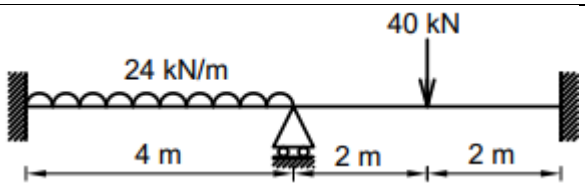
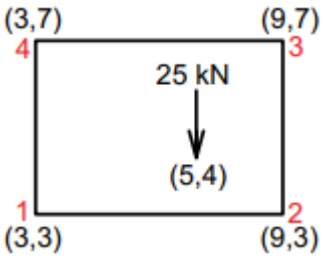
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**PE**
Course Name	**Finite Element Method of Analysis**	**Course Code**	**CIE 4065**
Semester	**VII**	**Max Marks**	**50**
Date	**29/11/2024**	**Duration**	**3 Hours**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Write the displacement model for the following elements: (i) Three noded bar element, (ii) two noded beam element, (iii) six noded triangular element	**3**	1	1,2,3,5,12	**2**
1B	Illustrate the procedure to obtain the transformation matrix for a two noded plane truss element.	**3**	3	1,2,3,5,12	**3**
1C	Determine the consistent load vector for the beam element shown in figure. 	**4**	4	1,2,3,5,12	**3**
2A	Evaluate the force developed in element 1 for the axially loaded structure shown in figure. Take E = 2 × 10⁷ kN/m². 	**3**	2	1,2,3,5,12	**4**

2B	Evaluate the displacements at nodes for the axially loaded structure shown in the figure. Consider E = 2 × 10^7 kN/m². 	**5**	2	1,2,3,5,12	**4**
2C	A space truss element of length 3 m is inclined at an angle such that $C_x = C_y = C_z = {}^{1}/_{\sqrt{3}}$. If the axial force developed in the element is 30 kN (tensile), determine the forces developed in the element in global direction.	**2**	3	1,2,3,5,12	**3**
3A	Evaluate the forces developed in the pin jointed structure shown in figure, if the horizontal and vertical displacements at node 2 are -6×10^{-6} m and -91.33×10^{-6} m respectively. Take E = 2 × 10^7 kN/m², A = 0.25 m². 	**5**	3	1,2,3,5,12	**4**
3B	Determine the rotations in the joints for the rigid jointed structure shown in figure, neglecting the axial and shear deformation, with EI constant throughout. 	**5**	4	1,2,3,5,12	**3**
4A	Analyse the beam shown in figure and calculate the forces in the first element carrying UDL. Consider constant EI.	**6**	4	1,2,3,5,12	**3**

					
4B	Determine the natural coordinates for a constant strain triangular (CST) element at its nodes and center.	**4**	5	1,2,3,5,12	**3**
5A	Illustrate the procedure to obtain the stiffness matrix for a two noded beam element using the principle of minimum potential energy.	**6**	4	1,2,3,5,12	**3**
5B	Determine the equivalent nodal load vector for the quadrilateral element shown in figure. 	**4**	5	1,2,3,5,12	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	*3*
2	*8*
3	*10*
4	*21*
5	*8*

BL	Total Marks
2	**3**
3	**34**
4	**13**
5	**0**